

Student Knowledge Assessment

CESE323 ERC1.0 - Cloud Security Envisioning & Strategy

Student Version - No Answers Included

Assessment purpose

This assessment verifies learner understanding of cloud security strategy, Microsoft Defender for Cloud posture management, workload protection selection, identity controls, AI workload security, and safe operational readiness practices for live or lab environments.

Course code	CESE323 ERC1.0
Course title	Cloud Security Envisioning & Strategy
Assessment type	Knowledge assessment - mixed format
Recommended duration	60 to 75 minutes
Total marks	100
Recommended pass mark	70% or facilitator-defined threshold

Student name		Date	
Email / ID		Cohort	
Instructor		Delivery mode	
Selected workload track 1		Selected workload track 2	
Assessor score	____ / 100	Result	Competent / Not Yet Competent

Assessment Instructions

- Answer all sections. Read each question carefully before selecting or writing your response.
- This is the student assessment. It intentionally does not include the answer key or marking memorandum.
- Use practical, security-first reasoning. Where questions involve a live environment, prioritize least privilege, written approval, scope control, auditability, rollback, and cost governance.
- Do not perform changes in a live customer or production environment unless the instructor has explicitly authorized the activity in writing.
- Do not use confidential, personal, financial, HR, legal, production, or customer data in examples, prompts, uploads, screenshots, or evidence submissions.

Assessment Blueprint

Domain	Assessment focus	Bloom level	Marks	Evidence type
Strategic initiation and governance	Security goals, risk vectors, compliance alignment, cloud security strategy	Understand / Apply	15	MCQ, short response
Unified visibility and Defender operations	Defender portal, XDR, incidents, alerts, Secure Score, recommendations	Understand / Apply	20	MCQ, matching, short response
CSPM and exposure reduction	Misconfigurations, over-privileged identities, attack paths, exposure management	Apply / Analyze	20	MCQ, scenario
Workload protection	Defender for Storage, Containers, App Service, AI Services and plan selection	Apply / Analyze	25	MCQ, scenario
Operational readiness	Permissions, live environment controls, data handling, rollback, cleanup, cost governance	Analyze / Evaluate	20	Short response, scenario

Source Alignment

This assessment is aligned to: the CESE323 workshop objectives, the four-part one-day workshop flow, Microsoft Defender for Cloud posture and workload protection topics, AI workload security, and the student readiness requirements for Azure access, permissions, safe lab usage, and live environment controls.

Student Answer Sheet

How to use this page

Use this answer sheet for Sections A and B. Written responses for Sections C and D must be completed in the response spaces provided after each question.

Section A - Multiple Choice Answer Sheet

Q1	A / B / C / D	Q2	A / B / C / D	Q3	A / B / C / D
Q4	A / B / C / D	Q5	A / B / C / D	Q6	A / B / C / D
Q7	A / B / C / D	Q8	A / B / C / D	Q9	A / B / C / D
Q10	A / B / C / D	Q11	A / B / C / D	Q12	A / B / C / D
Q13	A / B / C / D	Q14	A / B / C / D	Q15	A / B / C / D

Section B - Matching Answer Sheet

Item 1	Letter: ____	Item 2	Letter: ____
Item 3	Letter: ____	Item 4	Letter: ____
Item 5	Letter: ____	Item 6	Letter: ____
Item 7	Letter: ____	Item 8	Letter: ____
Item 9	Letter: ____	Item 10	Letter: ____

Section A - Multiple Choice Questions

Instructions: Select the best answer for each question. Each question is worth 2 marks. Total: 30 marks.

1. Which statement best describes the main purpose of Cloud Security Posture Management (CSPM) in Microsoft Defender for Cloud? [2 marks]

- A. To replace all network firewalls with endpoint agents
- B. To identify, prioritize, and remediate misconfigurations and posture risks across cloud resources
- C. To provide email filtering and anti-phishing only
- D. To back up production databases automatically

2. A customer wants visibility across Azure, hybrid, and multicloud resources before enabling workload-specific protection. Which approach is most appropriate first? [2 marks]

- A. Start with asset inventory, Secure Score, recommendations, and scope validation
- B. Enable every paid Defender plan across all subscriptions immediately
- C. Disable MFA temporarily to reduce access friction
- D. Move all resources into one shared resource group

3. Why should named user accounts be used during labs and live reviews? [2 marks]

- A. They reduce the need for RBAC
- B. They improve auditability and accountability for actions taken in the environment
- C. They automatically lower Microsoft Defender licensing costs
- D. They prevent all accidental changes

4. Which principle should guide role assignment for students during hands-on labs? [2 marks]

- A. Assign Global Administrator to every learner
- B. Use least privilege and grant only the access required for the assigned lab scope
- C. Disable Conditional Access to simplify the exercise
- D. Share one administrator account across the class

5. Which Defender capability is most directly associated with reducing risk from over-privileged identities and attack paths? [2 marks]

- A. Security Exposure Management and CSPM recommendations
- B. Azure billing export only
- C. Microsoft Word protected view
- D. DNS domain registration

6. A storage account lab requires uploads and suspicious access demonstrations. Which workload plan is the most relevant? [2 marks]

- A. Defender for Storage
- B. Defender for Containers
- C. Defender for App Service
- D. Defender for Endpoint only

7. A Kubernetes workload running on AKS is the focus of a workload hardening lab. Which Defender plan is most relevant? [2 marks]

- A. Defender for Storage
- B. Defender for Containers
- C. Defender for Key Vault only
- D. Defender for Office 365 only

8. A cloud-native web application hosted on Azure App Service needs posture and runtime protection examples. Which workload plan best aligns to this scenario? [2 marks]

- A. Defender for App Service
- B. Defender for Storage
- C. Defender for External Attack Surface Management only
- D. Microsoft Purview Audit only

9. Which AI security risk should be explicitly considered when using generative AI services in a lab? [2 marks]

- A. Prompt injection and sensitive data leakage
- B. Hard drive defragmentation
- C. Printer spooler configuration
- D. Desktop wallpaper leakage

10. What should be done before enabling paid Defender workload protection plans in a customer environment? [2 marks]

- A. Obtain scope approval, budget approval, and a cleanup or rollback plan
- B. Enable all plans at tenant root without informing stakeholders
- C. Use real HR and finance data to improve realism
- D. Disable audit logging to reduce noise

11. Which portal is most associated with Defender XDR incident correlation and unified security operations? [2 marks]

- A. Microsoft Defender portal
- B. Microsoft Store
- C. Azure Pricing Calculator only
- D. Microsoft Forms

12. Which control helps ensure students are working in the correct Azure directory for labs? [2 marks]

- A. Confirm the selected Microsoft Entra tenant and subscription before beginning
- B. Rename the browser shortcut
- C. Use a public Wi-Fi hotspot only
- D. Open multiple production tenants simultaneously

13. Which statement best reflects safe handling of lab data? [2 marks]

- A. Use only synthetic or approved test data and avoid confidential, personal, or production data
- B. Upload customer data to make alerts more realistic
- C. Use payroll files because they contain useful sample data
- D. Prompt AI services with confidential source code to test leakage

14. In a live environment review, what is the safest default posture for production systems? [2 marks]

- A. Read-only review unless written approval exists for specific changes
- B. Full production modification by all learners
- C. Disable all security controls for faster exploration
- D. Delete unused resources without review

15. Which outcome best demonstrates operational readiness at the end of this course? [2 marks]

- A. Students can connect strategy to practical Defender configuration, risk remediation, and controlled live review
- B. Students memorize product names only
- C. Students learn to bypass security controls
- D. Students avoid using Microsoft security portals entirely

Section B - Matching Concepts

Instructions: Match each concept in Column A to the most accurate description in Column B. Write the correct letter on the answer sheet. Total: 10 marks.

Column A - Concept	Column B - Description
1. Microsoft Defender XDR	A. A planned method to reverse or disable a change if it causes cost, operational, or security issues.
2. Cloud Security Posture Management (CSPM)	B. Identity and access management platform used for users, groups, roles, authentication, and access control.
3. Cloud Workload Protection Platform (CWPP)	C. Protection capability focused on Kubernetes and containerized workload security.
4. Microsoft Entra ID	D. Unified security operations experience for incidents, alerts, investigation, and response across Microsoft security products.
5. Conditional Access	E. Data governance, compliance, information protection, eDiscovery, audit, and privacy-related capabilities.
6. Microsoft Purview	F. Risk-based visibility and hardening guidance for cloud resources, misconfigurations, compliance, and attack paths.
7. Secure Score	G. Policy-based access decision capability using conditions such as user, device, location, risk, and application.
8. Defender for Containers	H. Runtime and workload-specific protection plans for servers, storage, databases, containers, app services, and AI services.
9. Defender for AI Services	I. Threat protection and posture visibility for AI services, including AI workload alerts and monitoring.
10. Rollback plan	J. A measurable indicator that helps prioritize security posture improvement over time.

Section C - Short Answer Questions

Instructions: Provide concise, technically accurate answers. Each question is worth 5 marks. Total: 30 marks.

1. Explain how the shared responsibility model influences cloud security decisions in Azure and Microsoft Defender for Cloud. [5 marks]

2. Describe a least-privilege role assignment model for a one-day Defender for Cloud lab where students need to observe posture and complete limited remediation tasks. [5 marks]

3. Explain how Secure Score, recommendations, and attack path analysis can help prioritize remediation work. [5 marks]

4. Compare two workload protection plans that could be selected for the course and explain when each one would be appropriate. [5 marks]

5. Describe two security risks unique or especially relevant to AI workloads, and explain how monitoring or governance can reduce those risks. [5 marks]

6. List five controls that should be in place before activating Defender for Cloud or changing security settings in a live customer environment. [5 marks]

Section D - Scenario-Based Applied Questions

Instructions: Analyze each scenario and provide a structured response. Each scenario is worth 10 marks. Total: 30 marks.

Scenario 1 - Posture and Exposure Prioritization

Scenario

A mid-sized organization has multiple Azure subscriptions, several over-privileged identities, inconsistent resource tagging, and a growing number of internet-facing services. The security team wants a practical way to identify the most important posture risks and communicate remediation priorities to leadership.

Required response: Answer all prompts below. [10 marks]

1. Identify the Defender for Cloud capabilities you would use first and explain why.
2. Describe how you would prioritize misconfigurations, identities, and attack paths.
3. Explain what evidence or metrics you would use to communicate progress to stakeholders.
4. State one risk of remediating too broadly without defined scope or approval.

Scenario 2 - Workload Protection Selection

Scenario

A customer runs a public web application on Azure App Service, stores uploaded documents in Azure Storage, and is piloting an Azure AI / Microsoft Foundry solution for support queries. The workshop allows only two live workload protection modules due to time and budget constraints.

Required response: Answer all prompts below. [10 marks]

5. Select two workload protection modules and justify your selection based on risk and business value.
6. Explain which workload you would demonstrate only as a guided walkthrough or instructor demo, if any.
7. List the minimum lab controls needed to avoid exposing confidential data.
8. Describe the cleanup actions that should occur after the assessment or lab activity.

Scenario 3 - Live Environment Review Controls

Scenario

During the final part of the workshop, a customer asks the class to activate Defender for Cloud in a production subscription and enable multiple workload plans immediately. Some students only have basic Azure experience, and the customer has not confirmed budget approval or a rollback plan.

Required response: Answer all prompts below. [10 marks]

9. Explain whether the requested activity should proceed and provide your rationale.
10. Define the minimum approval, scope, role, and change-control requirements before proceeding.
11. Recommend a safer alternative that still demonstrates operational value.
12. Identify how auditability, cost control, and data handling should be managed.

Student Declaration

- I confirm that the responses submitted are my own work unless collaborative work was explicitly authorized by the instructor.
- I understand that I must not use confidential, personal, financial, HR, legal, customer, or production data in this assessment.
- I understand that changes to lab or live environments must follow the instructor-approved scope and must not be made in production without explicit written approval.

Student signature: _____	Date: _____
Instructor / assessor signature: _____	Date: _____
Assessment decision: _____	Competent / Not Yet Competent: _____

Assessor Use Only

Section	Maximum marks	Score awarded	Assessor notes
Section A - Multiple Choice	30	_____	
Section B - Matching	10	_____	
Section C - Short Answer	30	_____	
Section D - Scenarios	30	_____	
Total	100	_____	

References and Course Alignment

- CESE323 ERC1.0 - Cloud Security Envisioning & Strategy Workshop Syllabus.
- CESE323 Student Pre-Course Readiness Requirements.
- Microsoft Learn collection for Cloud Security Envisioning & Strategy Workshop Syllabus.
- Microsoft Defender for Cloud, Microsoft Defender XDR, Microsoft Entra ID, Microsoft Purview, and Microsoft Foundry / AI security learning modules referenced in the CESE323 course collection.